Docking Analysis of Human Cytochrome P450 3A4

in complex with Protoporphyrin IX containing Fe (HEM)

(PDB ID: 1W0E)

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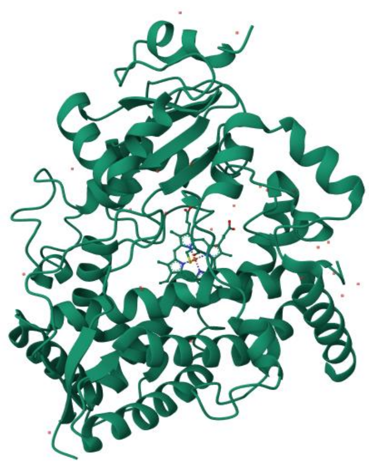
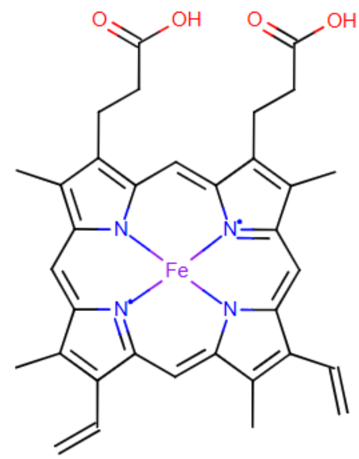
Introduction

Cytochrome P450 3A4 (CYP3A4), enzymes that form the central part of the body’s detoxification systems, are the first line of defense when it comes to eliminating unusual chemicals, such as drugs, poisonous plant compounds, carcinogens formed during cooking, and environmental pollutants. It is located in the human liver and is one of the most important members of the cytochrome P450 family. CYP3A4 is responsible for the oxidative metabolism of over 90% of marketed drugs, processing more drug molecules than any other cytochrome P450 isoform.

The crystal structure of CYP3A4 (pdbid: 1w0e) reveals a small active site that undergoes minimal conformational changes upon ligand binding. A key feature of CYP3A4 is its heme prosthetic group, which is essential for its enzymatic activity. This heme group enables the enzyme to catalyze the oxidation of substrates by incorporating one oxygen atom into the substrate while reducing the other to form water. This oxidation process is crucial for the detoxification and elimination of diverse compounds, ensuring the body's ability to effectively manage and clear potentially harmful substances. These structural and functional attributes of CYP3A4 make it of high significance in drug metabolism and chemical defense.

This docking study provides insights into the molecular interactions of CYP3A4 with its cofactor, protoporphyrin IX containing Fe (HEM). Docking simulations were carried out using AutoDock, and Chimera was employed for molecular visualization and analysis, providing detailed structural insights. Additionally, AutoDock Vina, recognized for its efficiency and speed, was used to complement and validate the docking results, offering an alternative perspective through its distinct scoring approach.





Crystal structure of human cytochrome P450 3A4

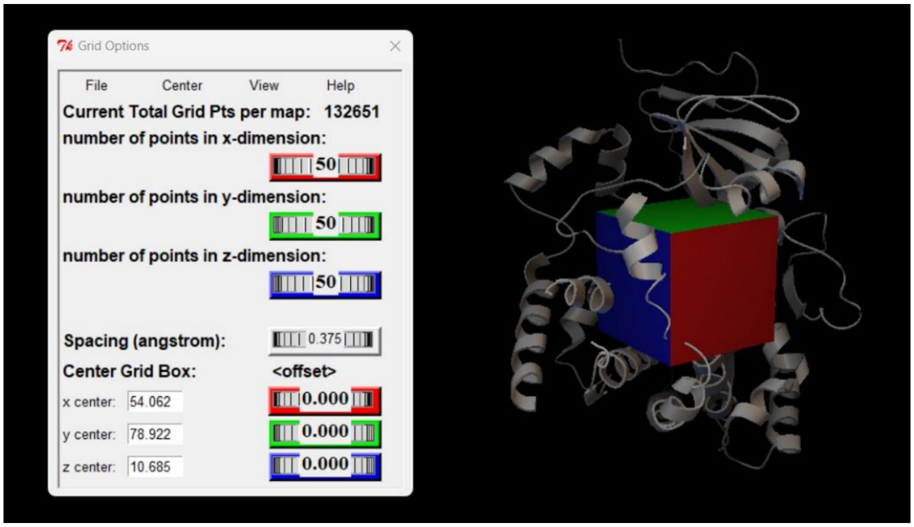
2D figure of the ligand (HEM)

AutoDock (ADT)

1. Parameters

The grid box for the docking calculations was set up by first loading the receptor protein and centering the grid on the ligand. The docking grid is defined and configured to fully encompass the receptor's binding site and the ligand at its larger extent. The grid parameters are as follows:

- **Grid Dimensions**: 50x50x50 pts, providing sufficient coverage for the binding pocket.
- **Grid Centering**: 58.562, 77.922, 14.185 (x, y, z).
- **Grid Spacing**: 0.375 Å per grid point.
- **Atom Types Considered**: A, C, Fe, OA, and N (from the ligand).



The ligand was prepared for docking with 8 torsional degrees of freedom, all of which were set as active to allow full flexibility during docking. The torsional degrees of freedom are located between the following atom pairs:

```
Total charge on ligand                              =  -1.996 e
REMARK  8 active torsions:
REMARK  status: ('A' for Active; 'I' for Inactive)
REMARK    1  A    between atoms: C2A_3624  and  CAA_3628
REMARK    2  A    between atoms: CAA_3628  and  CBA_3629
REMARK    3  A    between atoms: CBA_3629  and  CGA_3630
REMARK    4  A    between atoms: C3B_3635  and  CAB_3638
REMARK    5  A    between atoms: C3C_3642  and  CAC_3645
REMARK    6  A    between atoms: C3D_3649  and  CAD_3652
REMARK    7  A    between atoms: CAD_3652  and  CBD_3653
REMARK    8  A    between atoms: CBD_3653  and  CGD_3654
```

2. Results

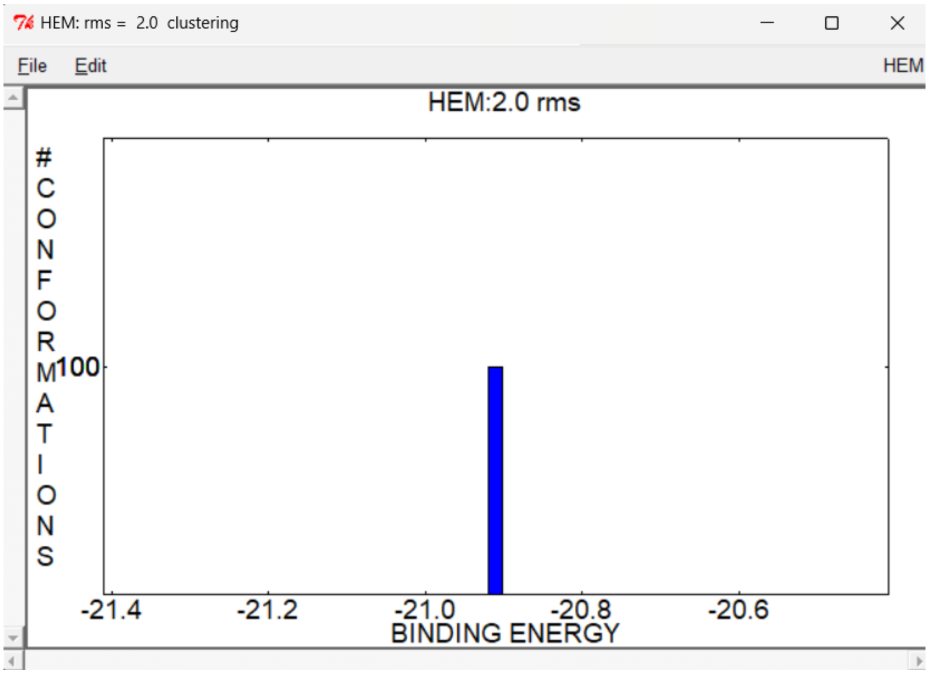
The loss of torsional free energy upon binding was estimated at +2.3864 kcal/mol, indicating the unfavorable entropy change due to the ligand losing flexibility as it explored the binding site. The final estimated free energy of binding was calculated as -20.36 kcal/mol, incorporating the following components:

- **Final intermolecular energy** of -22.75 kcal/mol.
- **Van der Waals, hydrogen bonding, and de-solvation energy** of -18.53 kcal/mol.
- **Electrostatic energy** of -4.22 kcal/mol.
- **Torsional free energy** of +2.39 kcal/mol.
- **Unbound system's energy** of -0.59 kcal/mol.

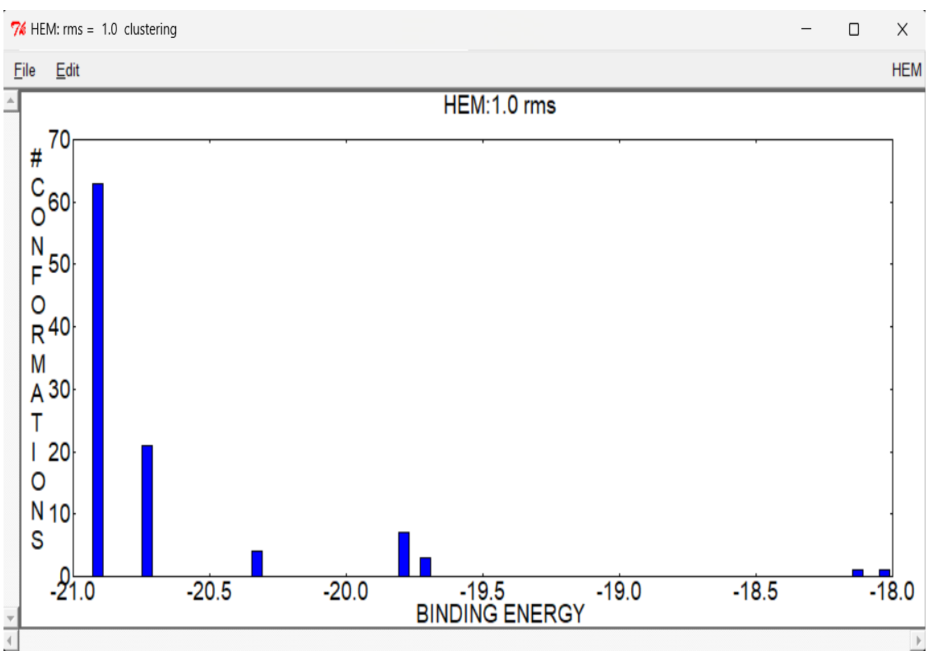
The docking calculations were performed using the Lamarckian Genetic Algorithm (LGA) with 100 independent runs, exploring the ligand's conformational space within the receptor binding pocket. The best estimated free energy binding (ΔG) was estimated at -20.36 kcal/mol, derived from LGA docking run #66, which yielded the lowest energy pose.

Clustering analysis of docking initially was performed with a RMSD tolerance of **2.0 Å** revealing a single cluster (RMSD = 0.00) across all 100 runs, which indicates a high degree of similarity among the docking results. The best cluster was observed at run #66, which had an RMSD of 0.00 and the highest binding energy (most negative), confirming it is the most stable conformation of the ligand in the binding pocket.

Re-clustering with a tighter tolerance of **1.0 Å** produced seven distinct clusters, suggesting additional potential binding conformations.



clustering histogram of the docking results for the ligand HEM using a root mean square deviation (RMSD) tolerance of 2.0 Å.



clustering histogram of the docking results with a root mean square deviation (RMSD) tolerance of 1.0 Å.

Total Clusters: 7 clusters obtained.

Cluster Distribution:

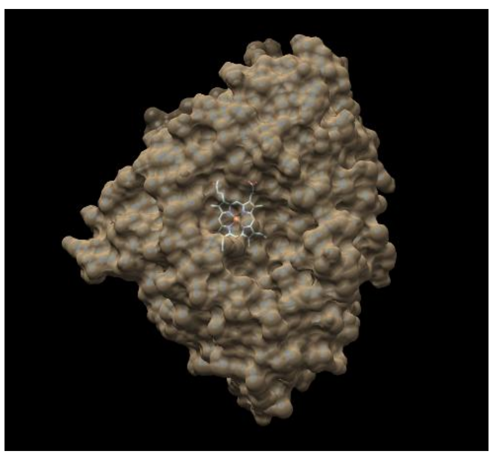
- **Cluster 1**: Contains 63 conformations.
- **Cluster 2**: Contains 21 conformations.
- **Cluster 3**: Contains 4 conformations.
- **Cluster 4**: Contains 7 conformations.
- **Cluster 5**: Contains 3 conformations.
- **Cluster 6 and Cluster 7**: Each contain 1 conformation.

The docking calculations were completed in 38 minutes and 35.20 seconds, with a CPU time of 15.80 seconds and a system time of 2.48 seconds. These results highlight the efficiency of the docking simulations in predicting the binding interactions between ligand and receptor.

3. Analysis

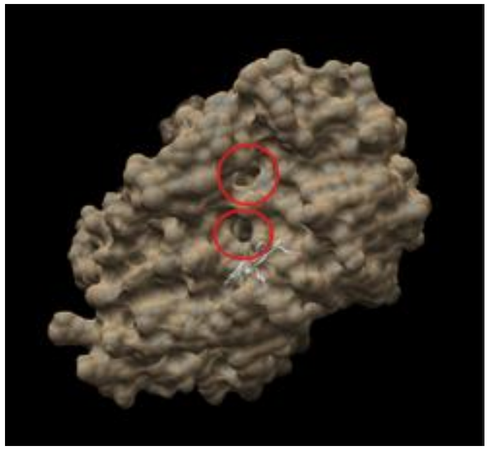
i. Molecular Surface (MS) Analysis

The top-docked pose of the HEM ligand fits well within the receptor's binding pocket, with no detected clashes between the ligand and the protein when evaluated using the "Find Clashes/Contacts tool in Chimera". The analysis was performed with specific parameters, including an allowed overlap of 0.6 and an H-bond overlap reduction of 0.4, which ensures the accuracy of the docking model. Additionally, contacts between atoms separated by fewer than four bonds were ignored, and intra-residue and intra-molecule contacts were excluded from the analysis. The results revealed zero clashes or unfavorable interactions, confirming that the ligand is properly accommodated in the binding site. This indicates that the top-ranked ligand pose does not induce steric strain and is likely to be stable in the receptor's binding pocket.



HEM ligand lies enclosed in the middle of the protein
(visualized using Chimera 1.18)

The channel like openings facilitate the binding of substrates like camphor and inhibitors such as carbon monoxide. They ensure that substrates can interact with the heme iron for oxygen activation, which is an important step for the catalytic addition of oxygen to the substrate.

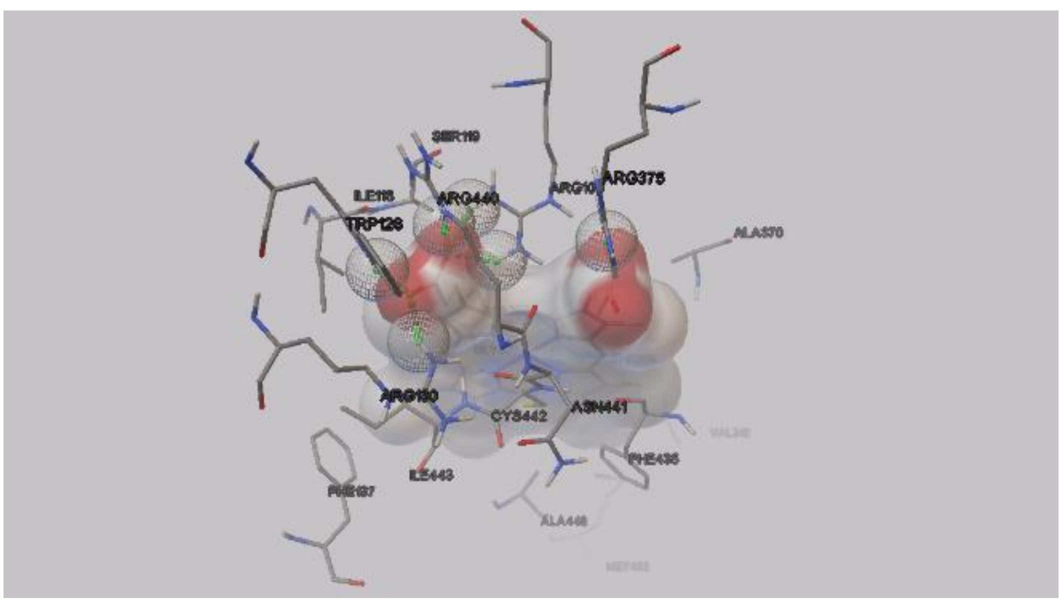


Channel like opening on top of the protein
(visualized using Chimera 1.18)

ii. Interaction Analysis

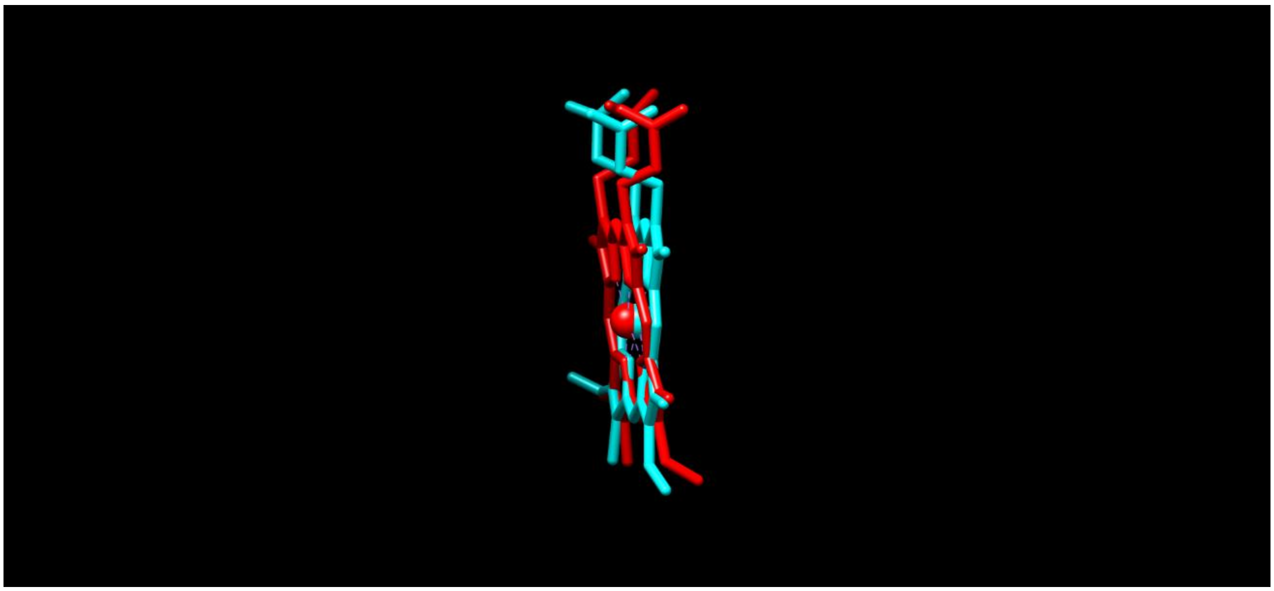
The interaction analysis of the top-docked ligand pose reveals hydrogen bonding interactions between the ligand and several amino acid residues within the receptor, which contributes to the stability and proper alignment of the ligand in the binding pocket. Six hydrogen bonds are formed, and they are as follows:

- **ARG105 (HH12):** side-chain nitrogen atom
- **ARG105 (HH22):** side chain nitrogen atom
- **TRP126 (HE1):** indole nitrogen atom
- **ARG130 (HH12):** arginine residue
- **ARG375 (HH12):** arginine residue
- **ARG440 (HE):** arginine residue



These interactions involve polar and charged groups and stabilize the ligand within the binding pocket. They also play a significant role in the receptor-ligand recognition and binding affinity.

iii. Structure Alignment



Docked model (red) vs Experimental (blue)

The figure shows high degrees of similarity between the docked ligand and the experimentally determined structure, with an RMSD of 0.940 Å calculated over 43 atom pairs (calculated using Chimera). This indicates that the docking procedure has accurately predicted the ligand's binding orientation and placement within the receptor's binding pocket. Despite this close alignment, minor differences in the precise positioning of the ligand are observed. This could be due to the flexibility of the ligand and receptor, which may not be fully captured in the docking model, or it may be due to the absence of solvent molecules or even protein flexibility.

The overall binding mode remains conserved, having key interactions between the ligand and receptor unchanged. Low RMSD value supports the reliability of the docking model, proving that the docked model provides a good approximation of the experimental structure.

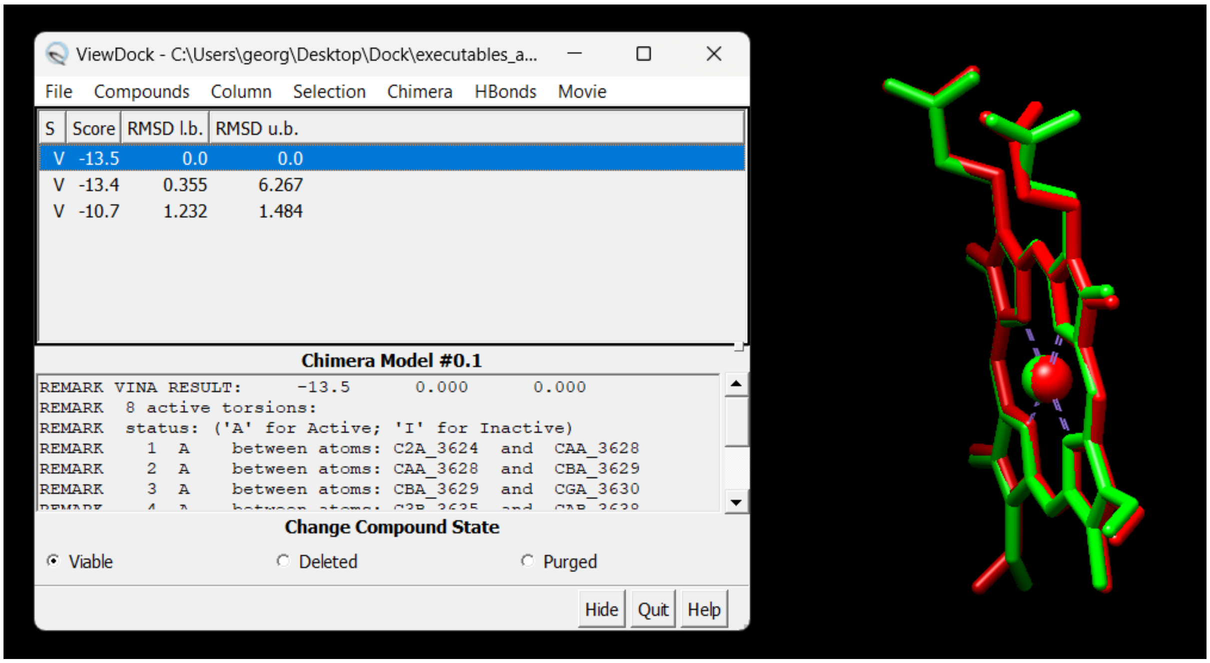
AutoDock Vina (ADT)

AutoDock Vina was used in order to verify and compare the docking results, adhering to the grid parameters used before. We obtained three different conformations. The following table shows the distinct ligand conformations with their binding affinities and RMSD values. These results confirm the accuracy of our initial findings using AutoDock.

Model	Affinity (kcal/mol)	dist from best model	
		rmsd l.b.	rmsd u.b.
1	-13.5	0.000	0.000
2	-13.4	0.355	6.267
3	-10.7	1.232	1.484

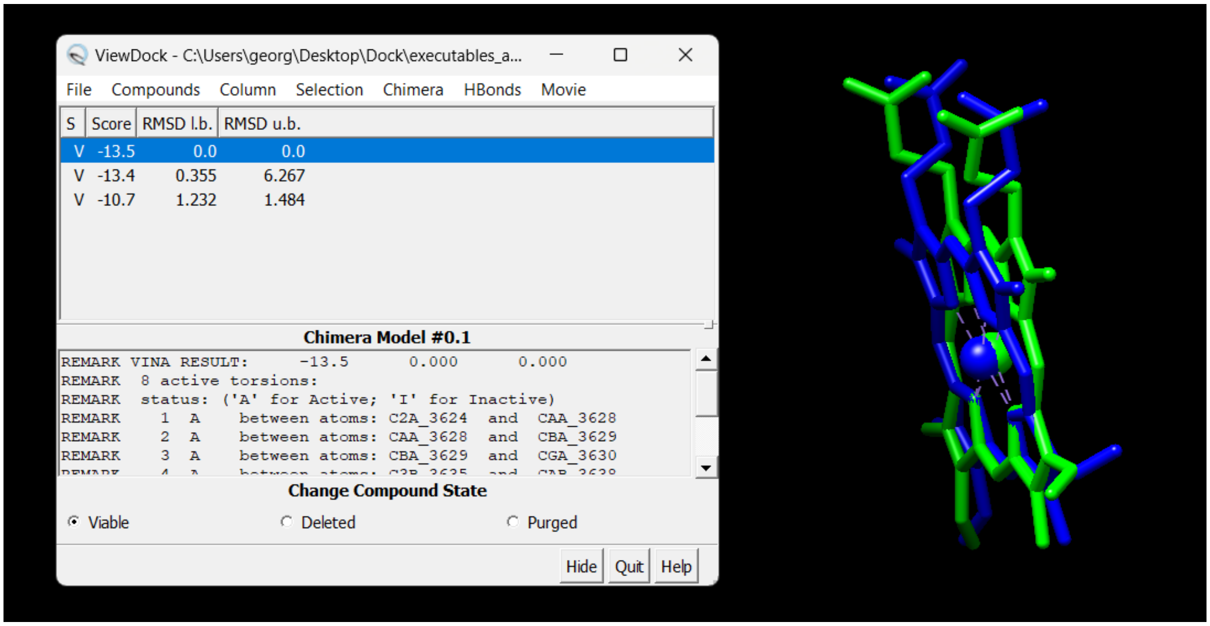
i. Model 1

Structure alignment using Chimera showed high similarity between the top conformations from each docking program and their alignment with the experimental one.



Model 1 (green) vs top conformation using AutoDock (red)

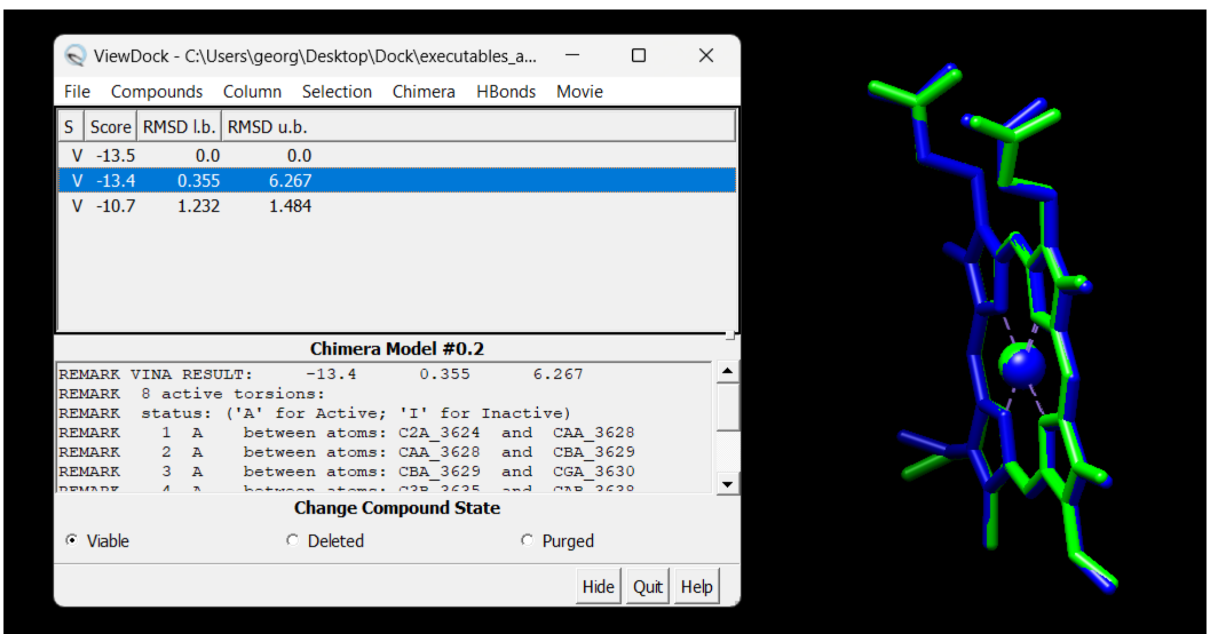
RMSD between 43 atom pairs is 0.642 angstroms



Model 1 (green) vs Experimental structure (blue)

RMSD between 43 atom pairs is 0.990 angstroms

ii. Model 2

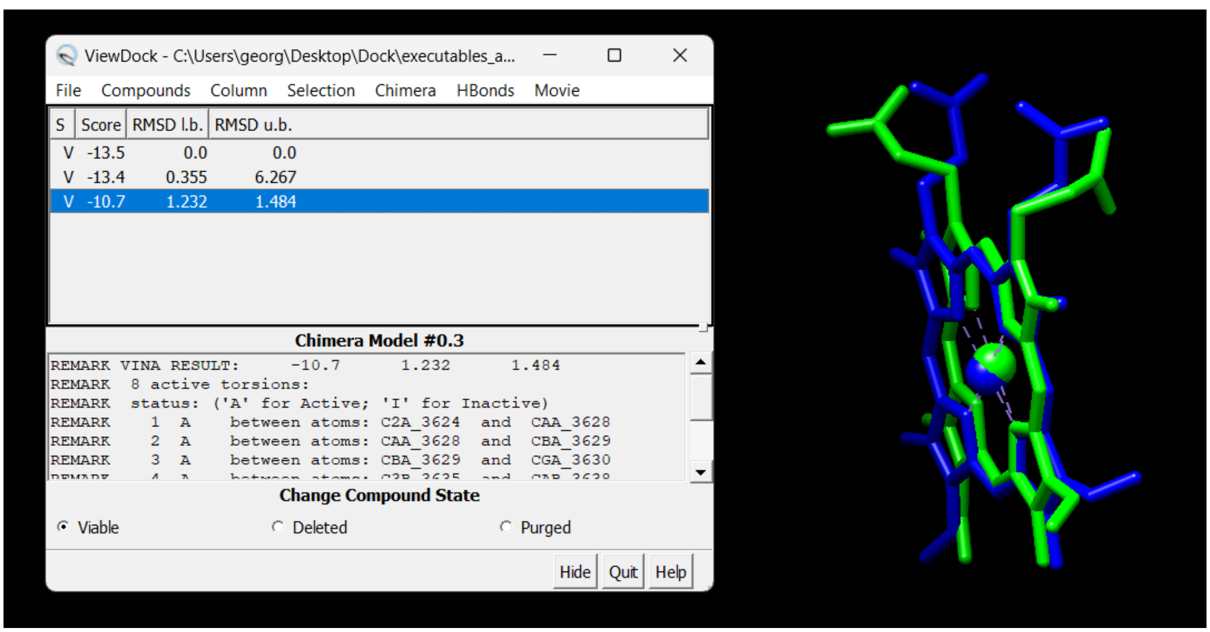


Model 2 (green) vs Experimental structure (blue)

RMSD between 43 atom pairs is 0.535 angstroms

The second best conformation from AutoDock Vina is closer to the experimental conformation. This may be due to the inherent flexibility of the docking process and the possibility that AutoDock Vina's scoring function, while effective, may not always perfectly rank conformations according to their binding affinity. Additionally, the conformations generated by the software are influenced by the starting coordinates, the search space, and the scoring algorithm. This conformation may therefore represent a more favorable or realistic interaction in this case, even if it is not ranked as the top result. It is also possible that the experimental conformation accounts for additional factors, such as solvent effects or protein dynamics, which may not be fully captured in the docking simulation.

iii. Model 3



Model 3 (green) vs Experimental structure (blue)

RMSD between 43 atom pairs is 1.298 angstroms

Conclusion

This study demonstrated how the enzyme Cytochrome P450 3A4 (CYP3A4) interacts with its ligand, Protoporphyrin IX containing Fe (HEM), using the protein model 1W0E. Using tools like AutoDock, AutoDock Vina, and Chimera for simulations and analysis, we found that the ligand fits well within the enzyme's active site. This research helps us better understand how CYP3A4 works with its ligand, which is important for future drug development and studying enzyme behavior.

Note:

The zip file will include the following:

1. "excecutables" folder containing files used for docking with AutoDock, prepared and run in AutoDock Tools 1.5.7 (ADT).

2. "executables_autodock_tools_vina" containing files used for docking with AutoDock Vina, prepared and run in ADT.

3. "executables_autodock_tools2" folder containing files used for our second docking run with AutoDock, prepared and ran in ADT files used for docking with AutoDock, prepared in ADT. As expected we observed minor differences in the Estimated Free Energy of Binding and different clustering results.

4. "executables_chimera_autodock_vina" folder files used for docking with AutoDock Vina, prepared and run in Chimera 1.18. Although the results from this preparation were not included in the report, this step was conducted to explore the use of alternative tools for docking preparation (ligand isolation, protein cleaning...) and analysis, as suggested in the assignment guidelines